

Appendix B

Broadband Matching

B.1 INTRODUCTION

Most students who have taken a course in fundamental antenna and transmission line theory consider themselves well-versed in matching an arbitrary load impedance Z_L into a resistive load. When challenged, they usually with some indignation will let it be known that they certainly know the workings of not only the single-stub tuner but also the double-stub tuner and perhaps even the triple stub. The sad facts are, however, that although all of these schemes will give us perfect match at *one* frequency, they will typically create a tremendous mismatch at others. In fact, it is sometimes stated even in reputable textbooks that the matching of an antenna always leads to a narrowband design; however, this is not necessarily so. In order to repudiate this notion, we shall in this appendix examine matching in general and, in particular, broadband matching. This has been necessitated by the realization that broadband matching is taught very superficially at most universities today, if it is mentioned at all.

This should not be interpreted as stating that broadband matching has been given insufficient time to mature. In fact, the author still remembers how he as a young engineer in the mid-1950s studied Chapter 3 in *Very High-Frequency Techniques* [119]. In his opinion, it still today remains the most authoritative text written on that subject. (He always made sure that his students in his first antenna course got a copy since the book, to the best of his knowledge, is out of print.)

Much of the broadband matching technique was developed during World War II for the purpose of countermeasures. One simply did not know where

in the frequency range the enemy could pop up. Further interest in broadband matching was fueled by the emerging TV industry. Ideally, the industry required transmitting antennas with $VSWR < 1.05$ over most of a TV channel (see also Section B.9.1). Expertise in that area appeared to have resided at a few antenna “houses.” They were in no particular hurry to share their knowledge with the rest of the world. (Industry sometimes exhibits a remarkable degree of modesty when it comes to products of commercial interest.) You simply had to go there and learn from the inside. The author himself joined the company “Rohde and Schwartz” in München, Germany, in the late 1950s. Beside being a well-known powerhouse in the area of measurement equipment (they were often compared to Hewlett Packard), they had an antenna department that probably was the most advanced in Europe in the VHF and UHF range at that time. Typically, they would match their antenna systems to have a $VSWR < 1.05$ not only in the operating TV channel but actually in an entire band—that is, such as 47–68 MHz and 174–224 MHz for the European Band I and Band III antenna systems, respectively. The author not only had the great fortune of participating in the design of several very challenging projects but also had several mentors, one in particular being Mr. L. Thomanek. When it came down to matching, his vision and experience were phenomenal.

Some papers were written on this subject. One in particular was by V. H. Rumsey and appeared in 1950 [124]. In spite of the fact that it was published later than reference 119, it mostly treated special cases as compared to the more general approach described in the earlier reference; however, there are indications that Rumsey actually did his work in England in the early part of World War II. Furthermore, reference 125 addresses primarily *amateurs*, but many *professionals* could benefit from reading it. Finally, Chapter 31 of reference 126 gives a general introduction and additional references on the subject of broadband matching.

B.2 MATCHING TOOLS

The fundamental tools used to match an arbitrary load impedance Z_L into another arbitrary impedance are shown schematically in Fig. B.1. At the very top we show from left to right:

- (a) Load impedance Z_L in series with a pure reactance jX .
- (b) Load impedance Y_L in parallel with a pure susceptance jB .
- (c) Transmission line with characteristic impedance Z_1 , length d , and termination in the load impedance Z_L .

Although some improvement in matching can be obtained by using lossy elements, it is in general not recommended since the possible reduction in mismatch loss in general is more than canceled by deterioration of the efficiency. Only at frequencies lower than ~ 100 MHz should this be considered since most of the noise at these frequencies comes from the outside through the antenna in the form

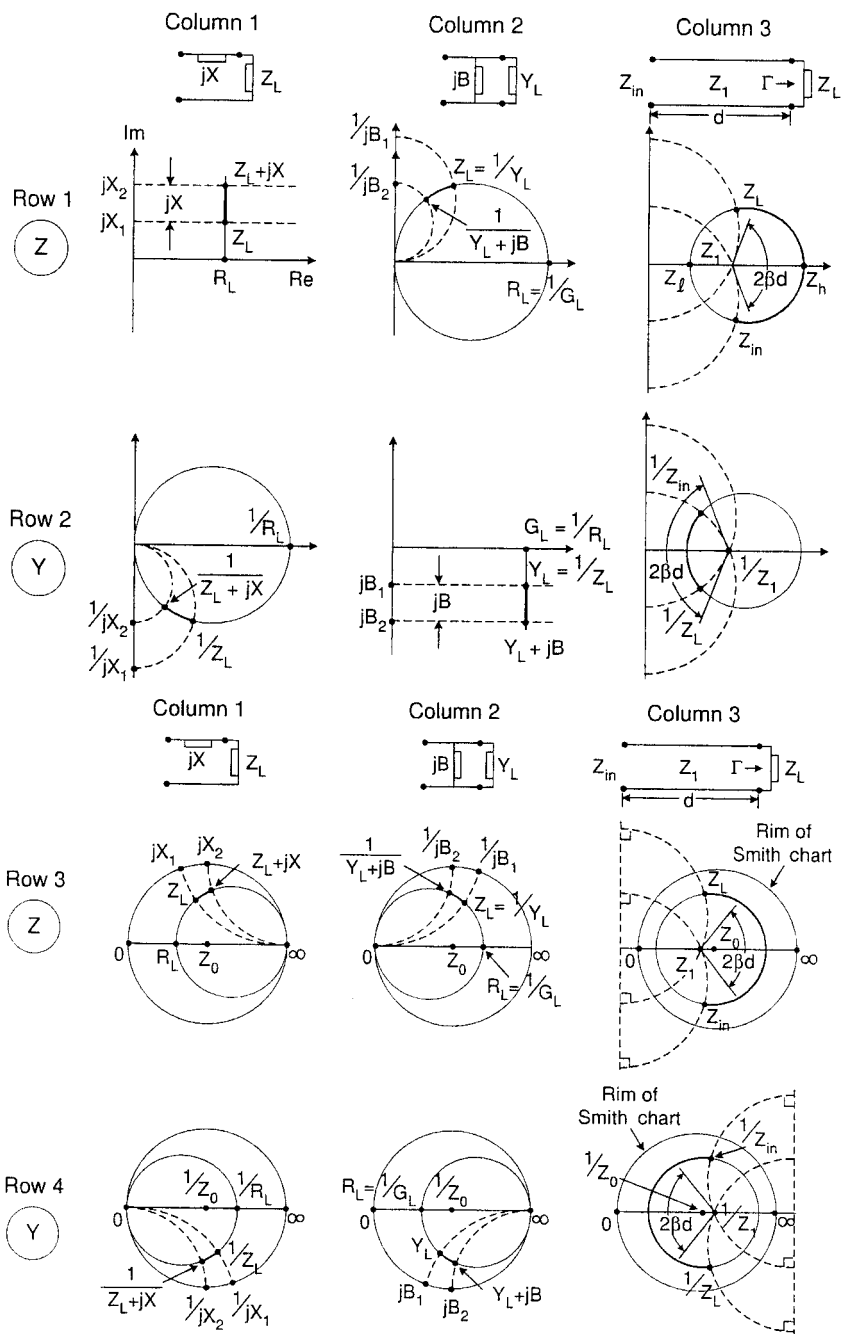


Fig. B.1 Basic tools for matching and their depiction in the complex plane: Pure reactances connected in series, column 1, or parallel, column 2, and a transmission line of length d and characteristic impedance Z_1 , column 3.

of Galactic or man-made noise. In other words, the signal and the noise are attenuated to the same degree, leaving the signal-to-noise ratio basically unchanged. Thus, since we are mostly interested in the general case, we shall in the following assume that all matching elements are idealized to be purely imaginary.

Figure B.1 is organized as follows. In row 1 we show the three cases depicted at the top in the impedance plane Z in rectangular coordinates. Next, in row 2 we show the same three cases in the admittance plane Y in rectangular coordinates. Similarly, we show in row 3 the impedance plane Z like row 1, but this time depicted in a Smith chart normalized to an arbitrary impedance Z_0 . Finally, row 4 shows the admittance plane Y , as in row 2, but depicted in a Smith chart normalized to an arbitrary admittance $Y_0 = 1/Z_0$.

More specifically, we show in column 1 what happened in any of the complex planes when we add a pure reactance jX in series with an arbitrary load impedance Z_L . As jX varies, we readily see how the series combination in row 1 moves along the vertical line going through Z_L . Furthermore, the admittance plot in row 2 is obtained from the case in row 1 by noticing that

$$Y = \frac{1}{Z}. \quad (\text{B.1})$$

Since (B.1) constitutes a bilinear transformation, we know from reference 123 that a general circle will be transformed into a general circle. In particular, we note that the real and imaginary axes will be transformed into themselves while the vertical line through Z_L in the Z plane will have its infinity point transformed into $(0, 0)$ in the Y plane and the point R_L into $1/R_L$. By further noting that the bilinear transformation is conformal [123], we see that the angle between the real axis and the locus in the Y plane must remain 90° ; that is, the center of the locus circle must be located on the real axis. That completely determines the locus circle in the Y plane as shown.

Finally, the horizontal lines $X = jX_1$ and jX_2 will have their infinity point in the Z plane transformed into point $(0, 0)$ in the Y plane while the two points jX_1 and jX_2 on the imaginary axis in the Z plane will be transformed into $1/jX_1$ and $1/jX_2$, respectively, on the imaginary axis in the Y plane. Again, noting that the bilinear transformation preserves angles we reason that the two horizontal lines $X = jX_1$ and jX_2 will be circles in the Y plane with their centers on the imaginary axis as shown. That completely determines the locus in the Y plane when adding a series reactance to an arbitrary impedance Z_L .

In row 3 we show the impedance plot from row 1 but now plotted in a Smith chart normalized to Z_0 . Again, we notice that the transformation from row 1 to row 3 is bilinear [123]; that is, general circles will be transformed into general circles. In particular, the imaginary axis in row 1 will be transferred into the rim of the Smith chart in row 3 while the real axis will be transformed into itself. Furthermore, the two lines $X = jX_1$ and jX_2 will have their infinity point transformed into ∞ in the Smith chart and the two points jX_1 and jX_2 are located on the rim of the Smith chart. We further note in row 1 that the angle between the two horizontal lines and the real axis is zero; that is, the center for

the circle arcs connecting ∞ and jX_1 and jX_2 must be located on the vertical line going through ∞ in the Smith chart. Finally, the vertical line going through R_L in row 1 is readily seen to be transformed into the circle going through ∞ and point R_L in the Smith Chart, row 3. That completely determines the locus in a Smith chart when adding a pure reactance jX in series with the arbitrary impedance Z_L . Finally the admittance case in row 4 is obtained as shown.

We next examine the case in column 2 of Fig. B.1 where we have connected a pure susceptance jB in parallel with the admittance $Y_L = 1/Z_L$. In that case it is most convenient to start with the admittance plane Y as shown in row 2. Adding a pure susceptance jB to the load admittance $Y_L = 1/Z_L$ will produce a locus that simply is the vertical line passing through Y_L . By arguing analogous to the case where we went from row 1 to row 2 in the series case (column 1), we readily observe that the locus in the impedance plane (Z of row 1) must be a circle with its center on the real axis and going through $(0, 0)$ and the point $R_L = 1/G_L$. Similarly, the two horizontal lines going through jB_1 and jB_2 in the admittance plane (row 2) will in the impedance plane (row 1) be two circles with centers on the imaginary axis and going through $(0, 0)$ and the two points $1/jB_1$ and $1/jB_2$, respectively, as shown. Finally, the Smith chart presentations in row 3 and row 4 are obtained by using the bilinear transformation features of the Smith chart transformation just as was done in column 1 when going from row 1 to 2 and from row 3 to 4, respectively.

Column 3, row 1, illustrates how to obtain the input impedance Z_{in} of a cable with characteristic impedance Z_1 , length d , and termination in the impedance Z_L . As explained in Appendix A, the input impedance Z_{in} will lie on a transformation circle going through the extreme points Z_h and Z_l on the real axis where

$$Z_h = Z_1 \frac{1 + |\Gamma|}{1 - |\Gamma|}, \quad Z_l = Z_1 \frac{1 - |\Gamma|}{1 + |\Gamma|} \quad (\text{B.2})$$

and the reflection coefficient

$$\Gamma = \frac{Z_L - Z_1}{Z_L + Z_1}. \quad (\text{B.3})$$

All the position circles will go through Z_1 and have their centers located on the imaginary axis [123]. Furthermore, the first is uniquely determined by also going through Z_L . We next draw the tangent to this position circle point through Z_1 . This is followed by drawing a second line through Z_1 , making an angle with respect to the first line equal to $2\beta d$ in the clockwise direction. The second line is a tangent to the position circle that determines the input impedance Z_{in} as shown.

The admittance case is shown in row 2. It follows directly from the impedance case in row 1 by the simple inversion $Y = 1/Z$, that is, circles will be transformed into circles. More specifically, the characteristic admittance is $Y_1 = 1/Z_1$ while the transformation circle is determined by the extreme points

$$Y_h = \frac{1}{Z_l} \quad \text{and} \quad Y_l = \frac{1}{Z_h}. \quad (\text{B.4})$$

The position circles are obtained completely analogous to the impedance case in row 1.

We next move to row 3 where we show the impedance plane but this time depicted in a Smith chart normalized to the arbitrary impedance Z_0 (i.e., $Z_0 \neq Z_1$ in general). Again, recalling that the Smith chart transformation from row 1 to row 3 is bilinear, we can immediately conclude that the transformation circles in row 1 will be transformed into similar circles in row 3. However, only if the Smith chart is normalized to Z_1 (i.e., $Z_1 = Z_0$) will the transformation circles in the Smith chart be concentric. The position circles are obtained completely analogous to the rectangular impedance case in row 1. Only if $Z_1 = Z_0$ will the position circle be straight lines going through the center of the Smith chart. Finally, the admittance case in row 4 is obtained from row 2 completely analogous to the impedance transformation above.

One might at this point wonder: “Why consider the general Smith chart case where $Z_1 \neq Z_0$?” Let us emphatically state that actual detailed calculations in a Smith chart should not in general be performed unless $Z_1 = Z_0$. However, as will be illustrated later, a typical situation occurs when Z_L is not a single point in the Smith chart but actually a given curve obtained by measurements (or otherwise) and plotted in a Smith chart normalized to Z_0 . It is in that case quite convenient to quickly draw the transformation circle for the extreme points of Z_L for various test values of Z_1 . See Appendix A. As discussed later, that will tell us what the smallest possible VSWR can be for a given curve for Z_L . See Section B.6.

One might be tempted to assume that the presentation above terminates our education in matching. Actually, it is just about to begin. The real challenge is to combine any of the three tools above in a design that transforms the arbitrary impedance curve Z_L into a small cluster of impedance points located anywhere in the complex plane. There is in general no simple unique approach. In the author’s opinion the best approach appears to be the graphical approach, as will now be illustrated with examples. (Oh yes, I admit for once, a little bit of experience is helpful as well.)

B.3 EXAMPLE: THE SINGLE SERIES STUB TUNER (NOT RECOMMENDED FOR BROADBAND APPLICATIONS)

Given: The load impedance Z_L as a function of frequency is shown in a Smith chart normalized to both 200 ohms and 100 ohms as shown in Fig. B.2. We are required to design a single series stub tuner as shown in the insert that transforms Z_L at the center frequency $f = 250\text{MHz}$ into 50 ohms. Based on that design at the center frequency we are next required to find the input impedance Z_{in} at the other frequencies.

Customarily one chooses the characteristic impedance Z_0 of the transmission line to be equal to the match impedance—that is, 50 ohms in our case. However, to illustrate a more general case for this type of matching we choose to use a characteristic impedance equal to 100 ohms. Thus we start with Z_L normalized

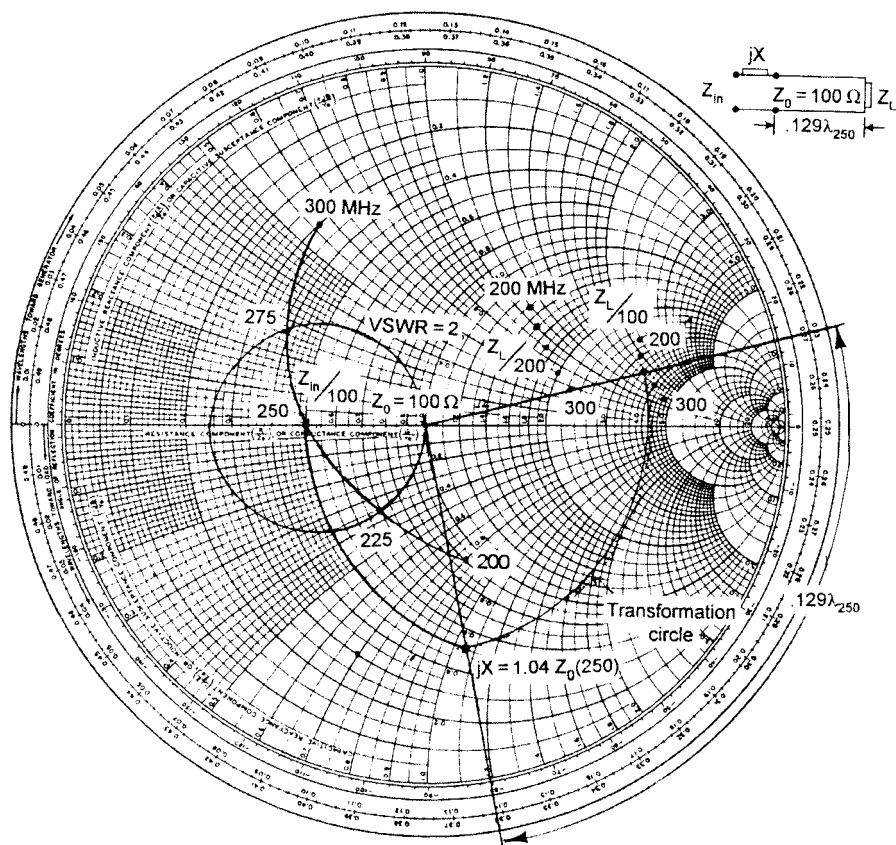


Fig. B.2 A load impedance Z_L is matched at the center frequency (250 MHz) to 50 ohms. The impedance Z_{in} is next obtained at the other frequencies as explained in the text. Not recommended for broadband applications.

to 100 ohms and rotate the value at $f = 250$ MHz until we intersect the constant resistance circle that goes through $\frac{1}{2} \cdot 100 = 50$ ohms (see Fig. B.1, column 1). As indicated in Fig. B.2, at the rim of the Smith chart this equals $0.129\lambda_{250}$. Finally, adding an inductance $jX = 1.04Z_0$ will bring the center frequency to $0.5Z_0 = 50$ ohms.

The other frequencies are determined by noting that their rotation in the Smith chart is proportional to the frequency—that is, $0.129 \cdot [(f/250)\lambda]$. Furthermore, we shall assume that the series reactance jX varies with frequency like an ideal lumped load; that is, it is also proportional to the frequency (stubs of transmission lines do not behave quite that simply). The final result is seen in the Smith chart normalized to $Z_0 = 100$ in Fig. B.2. The VSWR circle corresponding to 2 with respect to 50 ohms is also shown (see Problem B.2). Needless to say, broadbanded it ain't!

B.4 EXAMPLE: BROADBAND MATCHING

We are next going to match the same load impedance Z_L as shown in Fig. B.2 earlier. However, this time we will use a more sophisticated matching. A schematic of the network in question is shown at the bottom of Fig. B.3. (An example of a practical execution is shown later in Fig. B.8.)

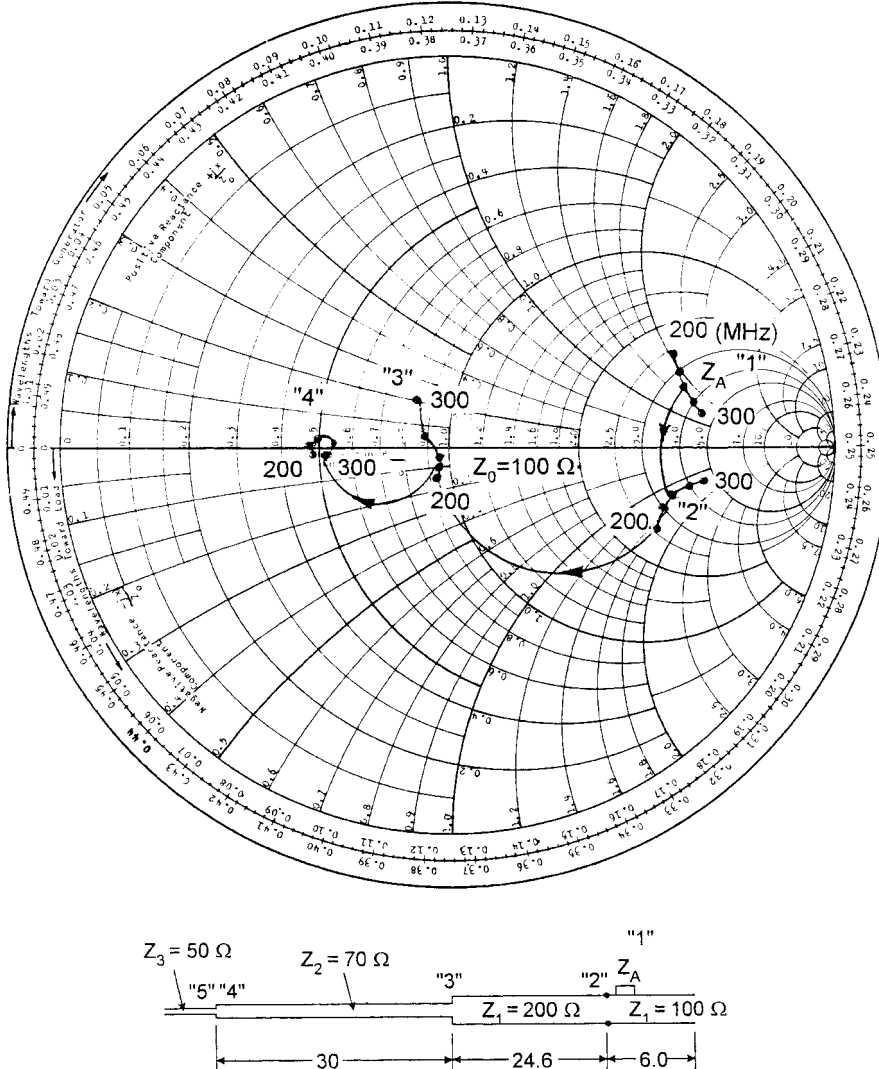


Fig. B.3 The same load impedance Z_L as given in Fig. B.2 but this time matched by a broadband matching scheme shown in the insert below. Curve 4 denotes the desired input impedance. Compared to the curve obtained in Fig. B.2, the improvement is quite noticeable.

Table B.1 Impedance Values for Fig. B.3

Frequency (MHz)	Z_A ("1") (ohms)	"2" (ohms)	"3" (ohms)	"4" (ohms)
200	$260 + j220$	$260 - j169.5$	$93.42 - j15.9$	$49.49 - j2.23$
225	$300 + j200$	$300 - j144.2$	$99.49 - j10.5$	$48.21 - j0.62$
250	$340 + j180$	$340 - j127.8$	$98.22 - j4.4$	$49.79 + j2.23$
275	$400 + j140$	$400 - j137.8$	$86.93 + j5.7$	$55.72 + j0.36$
300	$440 + j120$	$440 - j132.6$	$82.86 + j20.9$	$51.19 - j4.21$

We start calculating the input impedance Z_{stub} of the open-ended stub to the far right in the schematic. We have

$$Z_{\text{stub}} = j100 \cot \beta l_{\text{stub}}. \quad (\text{B.5})$$

When added in series with the load impedance Z_L curve 1, we obtain curve 2 (see Fig. B.1, column 1). We next transform curve 2 along a transmission line of length 24.6 cm and obtain curve 3, which again is transformed along a transmission line of length 30 cm yielding the final curve 4 (see Fig. B.1, column 3). Clearly, curve 4 in Fig. B.3 is far superior to the impedance curve in Fig. B.2 obtained earlier for a single series stub tuner (for details, see Table B.1). The obvious question is of course, Why the big difference?

The broadband concept was developed graphically. The fine-tuning was performed on the computer by Jonothan Pryor (like so many other items in this book). This would be a good place to start an optimization process.

B.5 THE "TRICKS"

When broadband matching, it is a fundamental rule that the original impedance curve 1 should as soon as possible be "reversed," which basically means that it should run the wrong way in the Smith chart—that is, counterclockwise. Statements to that effect will in general be met with strong objections, and they should! However, while it is true that an impedance curve will run clockwise as a function of frequency on the **average**, it is nevertheless possible for it to run backwards over a limited but still extensive frequency range. In fact, if we watch curve 2 in Fig. B.3 we observe that it does indeed run counterclockwise when watched from the center of the Smith chart (not if you watch it from a higher impedance to the right of the curve. Also note that curve 2 will run clockwise over a broader frequency range; that is, curve 2 merely has a loop).

The beauty of this situation becomes clear when we rotate curve 2 along the transmission line of length 24.6 cm. Since the rotations along the transformation circles are proportional to the frequency, the higher frequencies will travel further. However, since they are *behind* the lower frequencies (by virtue of reversing Z_L into curve 2), it becomes clear that curve 2 might be clustered together as they arrive at point 3. We observe this to be the case indeed; however, curve 3

has not landed at 50 ohms as desired and is also overcompensated (see later). For that reason we are inserting still another transmission line of length 30 cm and characteristic impedance 70 ohms. It serves two purposes: One is to transform curve 3 down to 50 ohms, the other is to obtain compression into the final impedance curve 4.

B.6 DISCUSSION

B.6.1 Overcompensation

Many questions remain to be answered. First of all, why don't we merely lower the characteristic impedance of the 24.6-cm transmission line from 200 ohms to, say, 160 ohms. That would land curve 3 closer to 50 ohms as illustrated in principle in Fig. B.4.

The typical problem with that suggestion might be that although the new transformation circles based on 160 ohms may appear in the figure to be as much apart from each other as the ones associated with 200 ohms, they might in effect lead to a curve 3 that is relatively wider than before because of the nature of the Smith chart (see also Problem B.2). But a more severe problem might be that the

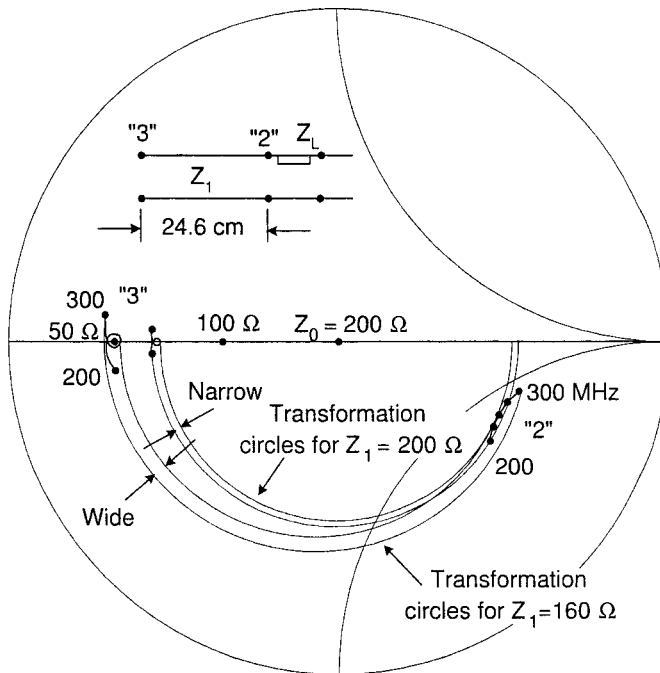


Fig. B.4 By changing the characteristic impedance Z_1 from 200 ohms to 160 ohms, we can land “3” closer to 50 ohms but the curve is wider and more stretched out than the original.
Note: Not to scale.

higher frequencies now are too far ahead of the lower frequencies as shown in somewhat exaggerated fashion in Fig. B.4. Thus, this overcompensation must be corrected either by using an additional transmission line as already demonstrated or by using other matching tools from Fig. B.1.

Alternatively, as shown in Fig. B.5, one might suggest to keep the characteristic impedance equal to 200 ohms and then instead move curve 2 somewhat outward from the center along the constant resistance circles as illustrated in Fig. B.1, row 3, column 1. This could be accomplished by increasing the reactance of the open-ended stub for example by making it slightly shorter. The limitation of this approach is that curve 2 eventually will start bending the wrong way, leading to a wider spacing between the transforming circles and ultimately leading to a wider curve 3. In addition, curve 3 might end up being overcompensated as we saw earlier in Fig. B.4.

The conclusion of the discussion above is that curve 2 should be manipulated to produce transforming circles as close to each other as possible. That is not likely to land curve 3 at the right impedance level. This can subsequently be corrected by use of one or more transformations as explained above. Alternatively, other tools from Fig. B.1 can be used as discussed next.

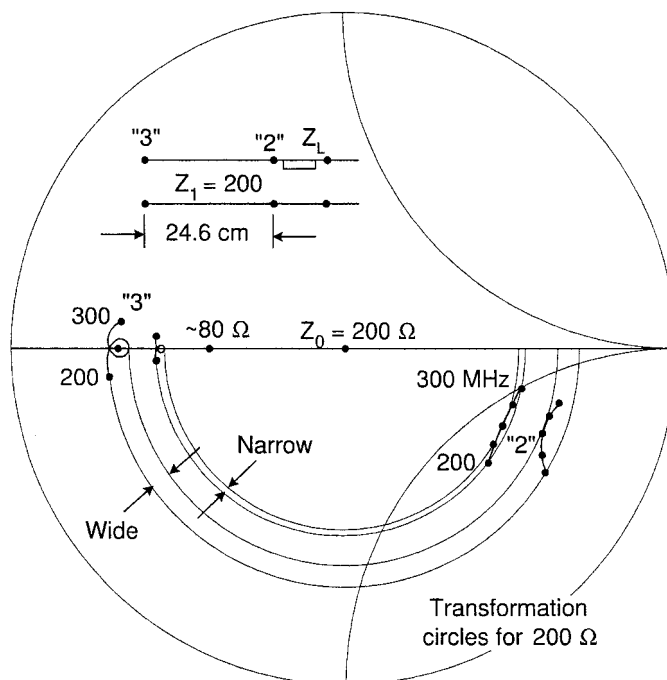


Fig. B.5 By increasing the reactance of the open stub to the far right in the schematic above, we typically bend curve "2" too much resulting in a new curve "3" closer to 50 ohms but too wide. Note: Not to scale.

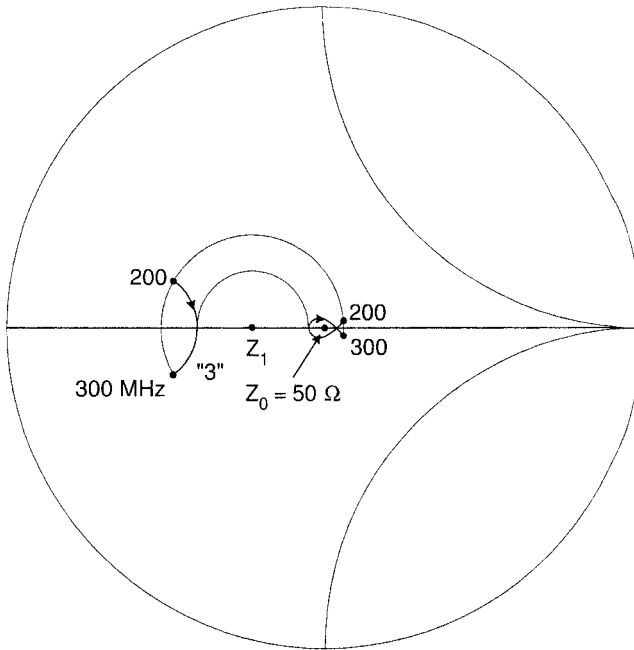


Fig. B.6 If the high frequencies are behind the low ones, the impedance curve can be compressed by transformation through a transmission line with high characteristic impedance Z_1 as shown. Note: Not to scale.

B.6.2 How to Correct for Undercompensation

We saw above an example where the higher frequencies were ahead of the lower (overcompensation). The opposite might also be the case as shown in Fig. B.6. To correct for this dilemma, we may transform curve 3 through a transmission line with a characteristic impedance Z_1 that is higher than curve 3. If that does not land us in the desired area, more subsequent transformations might be necessary.

B.6.3 Alternative Approaches

An undercompensated curve 3 can also be compressed by an insertion of a series circuit or equivalent as shown in Fig. B.7 to the left. At the lower frequencies the series circuit is capacitive, which will move the lower frequencies of curve 3 toward the capacitive part of the Smith chart along the constant resistance circles (see Fig. B.1, row 3, column 1).

Similarly, an overcompensated curve 3 can be treated by placing a parallel LC circuit in parallel with the impedance curve 3 as shown in Fig. B.7, right. At the lower frequencies the LC circuit is inductive, which will push the lower frequencies of impedance curve 3 up toward the inductive part of the Smith chart. They will move along the circles going through (0, 0) in the Smith chart as shown in Fig. B.1, row 3, column 2.

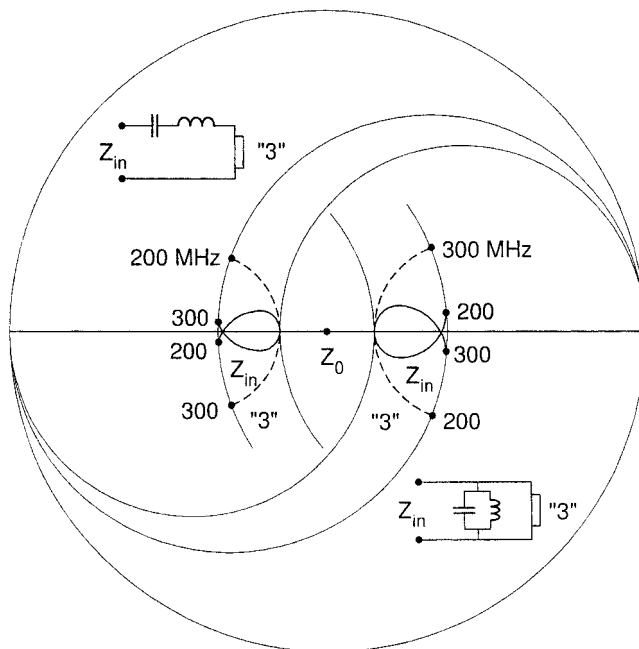


Fig. B.7 Undercompensation, shown to the left, can also be corrected by a series insertion of a series LC circuit or equivalent like an open-ended transmission line. Similarly, overcompensation, shown to the right, can be treated by insertion of a parallel circuit in parallel with "3".

B.7 ON THE LOAD IMPEDANCE Z_L

So far not much has been said about the load impedance Z_L . However, after a couple of tries into the fine art of broadband matching, it soon becomes clear that Z_L just cannot be any impedance willy-nilly. The fundamental discovery is that since all the matching tools shown in Fig. B.1 or combinations thereof varies with frequency in a monotonic and predictable fashion, so must Z_L in order to obtain a final impedance confined to a small area in the complex plane. Thus, Z_L should vary in a *smooth* fashion with frequency; that is, there should be no sudden accelerations or de-accelerations as a function of frequency.

Furthermore, Z_L itself should have as much bandwidth as possible. When working with wire elements, this is typically accomplished by using a large wire diameter (remember the "Fat is beautiful" doctrine).

The physical explanation for this fact is simply that by using thick elements we are displacing or reducing the energy stored around the elements and that translates directly into a larger bandwidth (see also Section 6.12.1.1 and Appendix D).

Furthermore, the curve depicting Z_L should be shaped or curved the proper way. More precisely, after it is reversed (for example) as illustrated above, it should ideally follow a transformation circle as close as possible. This turns out to be one of the more intriguing tasks in broadband matching. Typically, the

antenna in question might be a single row of dipoles mounted side by side in the H plane; that is, the mutual coupling is strong. Recall now that the leading term in the mutual coupling between two elements with interelement spacing d is

$$Z_m \sim C \frac{e^{-j\beta d}}{d} \quad (\text{B.6})$$

(C is a complex constant); that is, the mutual impedance can in principle be located anywhere in the complex plane. Consequently, when added to the self-impedance of a single element, it can for the proper spacing d actually bend the curve Z_L the “wrong” way in a limited band. The element spacing d must be chosen primarily for impedance reasons (typically around 0.5λ) and not for either maximum gain ($0.7\text{--}0.8\lambda$) or very small spacings as recommended in Chapter 6 for ultra-wide band but with higher VSWR. However, the author can recall an interesting case where six vertical dipoles each with their own separate reflector grids were mounted axially; that is, the mutual coupling was rather weak. In that case it was simply not possible to obtain an ideal Z_L . The problem was eventually solved in a somewhat unusual way. As stated above, each dipole was backed by an individual reflector grid made of four closely spaced vertical rods interconnected by four widely spaced horizontal runners. Normally, one must be careful that the grid openings are small enough that they do not start to resonate at the operating frequencies. In that event the overcoupled impedance from the grid openings can produce highly undesirable loops in the impedance curve for Z_L . In this case we did precisely that. By cutting away part of one of the horizontal runners, some reflector loops of twice the normal size were produced. As a result of that “undesirable” resonance, the curve for Z_L could be bent just enough to clear the spec (because we anticipated problems, we had agreed to a VSWR < 1.05 only over 11% bandwidth).

Finally, it should be pointed out that Z_L also might be adjusted to the proper magnitude. In fact, Z_L as given in the example above is a bit too high, leading to a characteristic impedance of the first transformer equal to 200 ohms. While this value is fine when working with twin leads, it leads in general to an inner conductor too thin to be practical when we use a coaxial cable as shown in Fig. B.8. This is in general a relatively easy to fix when we recall that while the radiation resistance of a $\lambda/2$ dipole varies relatively little with its diameter, a full-wave dipole has a radiation resistance amounting to several thousand ohms for thin wire radii while it might be as low as a couple of hundred ohms for large wire radii [127]. Although our dipole is neither a half- nor a full-wave dipole but more somewhere around 0.7λ , its radiation resistance can easily be manipulated by the wire radius as well.

We may conclude from the discussion above that quite a considerable time and effort is often spent to manipulate the load impedance to be just right, or as it is sometimes expressed in the matching jargon, “Make sure it is born right.” No question about it, experience plays a significant role in that development and there is no better way than to fight it out partly by hand in a Smith chart. Once a concept is developed, the fine-tuning can be done on the computer.

The load impedance Z_L is customarily obtained by measurement. In industry, nobody believes the calculations of thick dipoles except the guy who did them. Everybody believes measurements, except the guy who took them.

B.8 EXAMPLE OF A PRACTICAL EXECUTION

Many possibilities exist when reducing the matching scheme above to practice. One is shown in Fig. B.8. First of all the dipole has, as mentioned earlier, a total length of $\sim 0.7\lambda$. If we plot the voltage (or potential) along such a long element, we observe that it is zero somewhere in the middle of each dipole half (see Fig. B.8). We may support these two halves at the neutral point and essentially leave the current and voltage distribution on the dipole unchanged. In addition, we can feed the dipole through one of these support structures (such an arrangement belongs to a class of baluns called “natural baluns;” see reference 116).

We further note that the open-ended stub to the far right in the schematic is placed inside the right dipole half (the length can be adjusted by moving a sleeve with a tool inserted through a hole in the end of the right dipole half). We next follow with the transmission lines of lengths 24.6 and 30 cm, respectively. They are executed in the form of coaxial lines placed inside the left dipole half. Typically one might, as in this case, resort to folding one (or more) of the cable sections in order to obtain a length sufficiently long to accommodate the required transformations.

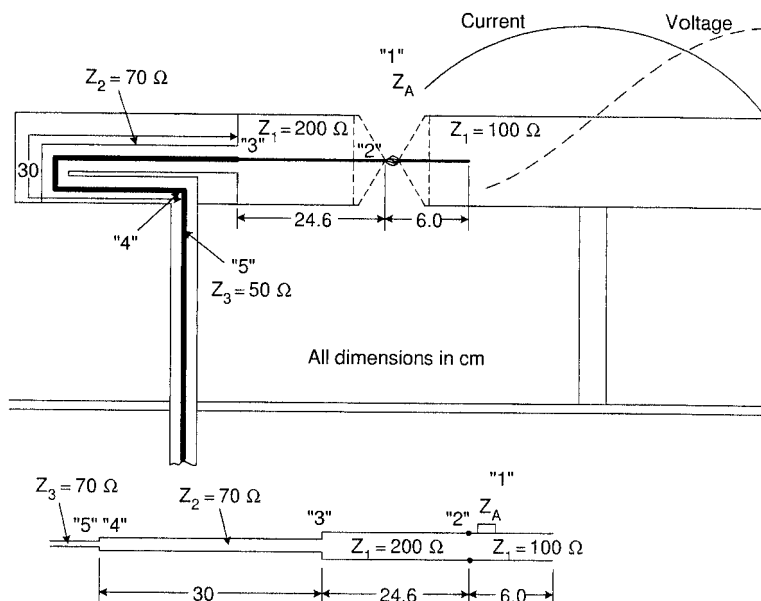


Fig. B.8 A practical execution of the broadband matching schematic shown in the insert. Consists of a dipole with total length $\sim 0.7\lambda$ and built-in matching network.

B.9 COMMON MISCONCEPTIONS

B.9.1 Should One Always Conjugate Match?

Many readers assume that two devices connected to each other should always be conjugate-matched. This is not necessarily the case. One should reflect upon the fact that conjugate matching means that the reactive parts of the two impedances simply cancel each other (in order to transfer maximum power) while the real parts are equal. As any fundamental textbook in circuit theory shows, this condition leads to maximum transfer of real power between the two impedances. What is not always pointed out is that since the two real parts are equal, the same amount of power will be lost in the generator resistance as is delivered to the receiver resistance. In other words, the efficiency is only 50% when we conjugate match. The reason for this omission is basically that nobody actually cares as long as we merely consider transfer of power at low levels as, for example, a receiving antenna connected to a receiving load. All we care about in that case is simply how to get maximum power into the receiver, not if some power is lost in the form of heat or reradiation in the process (there is no significant heat developed, nor is there a bill to be paid for energy lost or otherwise).

It is quite a different story if we work at high power levels. Take a typical power generator where we get paid for energy delivered to the users out there. If the combined resistance of these amounts to R_{user} and the generator resistance is denoted R_{gen} , then the power being paid for is $R_{\text{users}} I_{\text{gen}}^2$ while $R_{\text{gen}} I_{\text{gen}}^2$ represents loss that will not be paid for (at least not directly). Obviously it will be the aim of the generator designers to make R_{gen} as low as possible, and it is strongly recommended not even to attempt to conjugate match such a well-designed power generator. For one, you probably could not pay the bill and you most certainly would be arrested for attempted sabotage.

Admittedly, most readers of this book will in general have a limited interest in power generation (it is just a plug in the wall that is there when you need it). However, it becomes considerably more interesting if you must connect your antenna to a high-power transmitter. You will then find that the transmitter designers are quite interested in efficiency and as a result produce transmitters with internal impedances of just a few ohms. Any suggestion on your part of conjugate matching will be met with a stern lecture by the transmitter designers that the load connected to their transmitter better be just what they tell you and within rather narrow limits at that. In other words, a high-power transmitter can be somewhat temperamental and should not be tampered with (for one it is a nonlinear device and quite sophisticated).

As a result, there is in general a severe “mismatch” between a high-power transmitter and the antenna cable. The next question then becomes, “Is this of any relevance?:” The answer is, “Absolutely.” Take, for example, a TV transmitting antenna that is connected to a high-power transmitter via a long cable of length l . Denoting the reflection coefficients between the cable and the antenna for Γ_A and between the cable and transmitter for Γ_{Tr} we readily see that the

relative level of the first ghost is given by $\Gamma_A \Gamma_{Tr} e^{-2j\beta l}$. In order not to see this ghost, it should be sufficiently low, depending on the time delay through the cable. For a typical cable length of a couple of hundred meters, it has been found to be around 38 dB. Furthermore, because of the poor match between transmitter and cable, we typically have $\Gamma_{Tr} \sim 2$ dB. Also, the single way cable loss may be ~ 2 dB; that is, we must require Γ_A to be ~ 32 dB corresponding to a VSWR ~ 1.05 . Quite a responsibility on the frail shoulders of the antenna engineer. (For a receiving antenna with a much shorter cable the ghost level drops to typically ~ 26 dB. Furthermore, the reflection coefficient at the receiver with VSWR < 2 leads to $\Gamma_{Rec} < 10$ dB; that is, we must require a VSWR < 2 also for the antenna.)

But the most interesting situation for most readers is probably how to control the RCS of an antenna array. As explained in Chapter 2, Section 2.12, we must simply require the reflection coefficient between antenna and transmitter impedance to be as low as possible, which is in direct contradiction to the low transmitter impedance as discussed above. The problem is solved, for example, by use of circulators as explained in Section 2.12.2. An incident signal will in that case be perfectly absorbed by the load in port 3, while the transmitter independently is delivering power to the antenna without loss of power.

B.9.2 Can New Exotic Materials Be Churned Out by Running Backwards?

An impedance curve that runs counterclockwise in the Smith chart is likely to produce quite a commotion. In fact, this author is known for having thrown students out of his office on that account. However, one should be very careful to distinguish between an impedance curve that goes the wrong way at all frequencies and one that just exhibits a loop in a limited frequency range.¹ See also Section B.5.

If an impedance curve is shown over a reasonably large frequency range, a competent antenna engineer will usually size up the situation very quickly (here we have another good reason for plotting our impedance curve in the complex plane). However, if only a small frequency range is at hand, an inexperienced engineer or a computer could be misled into “thinking” we had discovered something new. The interpretation of this discovery appears to depend somewhat on one’s background. Engineers may think they have discovered negative inductances and capacitances while physicists may interpret their observations as a new material with negative μ and ϵ . Many wonder if backward running “materials” can be designed at optical frequencies. Answer: Certainly! How: Left as an exercise for the student. Does the “material” really have negative μ and ϵ ? Answer: No.

¹ This situation in no way violates Foster’s reaction theorem [101], which applies only to *lossless* reactances while antenna impedances of radiating antennas are (or should be) lossy to represent radiated power.

B.10 CONCLUDING REMARKS

We have presented three basic matching tools as shown in Fig. B.1. We illustrated how to use these tools in a case where we matched a given impedance first by a single stub tuner and next by a broadband matching scheme where the given impedance was inverted, which basically means that it was running counterclockwise in the Smith chart over a limited frequency range. The improvement in VSWR amounted to more than an order of magnitude.

This presentation makes no claim to be complete. It was only intended to introduce the reader to the concept. In fact, the number of combinations of the matching tools is actually infinite. And like good music, the real beauty comes out when the various voices are blended together.

PROBLEMS

B.1 Prove that

$$Z_1 = \sqrt{Z_l \cdot Z_h}$$

where Z_l and Z_h are given by (B.2).

B.2 Prove that the voltage standing wave ratio is determined by the extreme impedances Z_l and Z_h (B.2) as

$$\text{VSWR} = \sqrt{\frac{Z_h}{Z_l}}$$

B.3 Design a broadband matching network for the terminal impedance shown in Fig. 5.36, bottom. This can actually be done simpler than the example treated in Fig. B.3. You may “land” the matched impedance anywhere on the real axis between 150 and 260 ohms.